

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)			Indicative content Elements and water • reactivity increases down both groups • both potassium and calcium metals react with water to give the hydroxide and hydrogen • Group 1 metals react more vigorously • Group 1 metals lose only one electron while Group 2 metals lose two electrons / Group 1 metals form cations easier than Group 2 metals • potassium melts into a ball and catches fire/lilac flame • calcium produces a steady stream of bubbles/cloudy white solution forms • $2K + 2H_2O \rightarrow 2KOH + H_2$ • $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$ Carbonates • Group 1 and Group 2 carbonates show different properties • Group 1 carbonates are soluble, Group 2 carbonates are insoluble 5-6 marks Comparison of similarities/differences, description of reactions and appropriate equations <i>The candidate constructs a relevant, coherent and logically structured method including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.</i> 3-4 marks Comparison of similarities/differences and description of reactions OR description of reactions and appropriate equations <i>The candidate constructs a coherent account including most of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary are generally sound.</i>	4	2		6		1